# 9.1 Using Normal Distributions

Name _____ Period ____ Date _____

Algebra II 9.1 · Student sheet

**I can…**

- use the 68-95-99.7 rule to find areas under a normal curve.
- find probabilities for normally distributed data.
- tell whether data are approximately normal and how skew affects the mean and median.

**Vocabulary** · fill in the missing word

**Normal distribution** — a _____, symmetric distribution with mean μ and standard deviation σ

**Normal curve** — the bell-shaped graph of a normal _____; total area 1

**68-95-99.7 rule** — about 68%, 95%, and 99.7% of the data are within 1, 2, and 3 standard _____ of the mean

**Skewed** — not _____: the mean is pulled toward the tail

## Part 1 · Coin-flip bell

Flip 10 coins. Number of heads: _____. Record the class results.

| Heads | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Students | | | | | | | | | | | |

What shape is it? Where is the center? _____

## Part 2 · Notes: fill in the blanks

Within 1σ of the mean: _____% within 2σ: _____% within 3σ: _____%

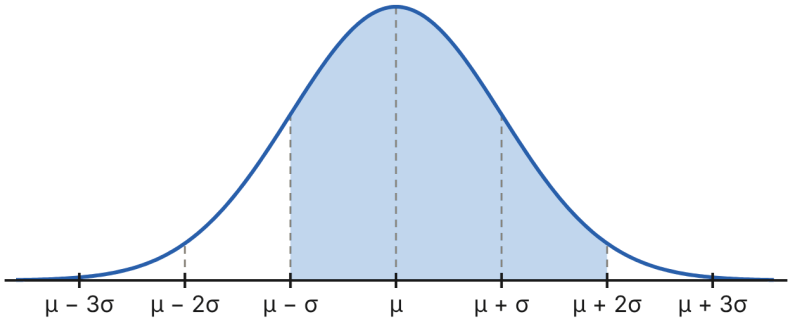
Bands on each side: μ to σ: _____%, σ to 2σ: _____%, 2σ to 3σ: _____%, beyond 3σ: _____%

Skewed right: mean _____ median. Skewed left: mean _____ median.

## Part 3 · Examples (work them with the class)

### Example 1

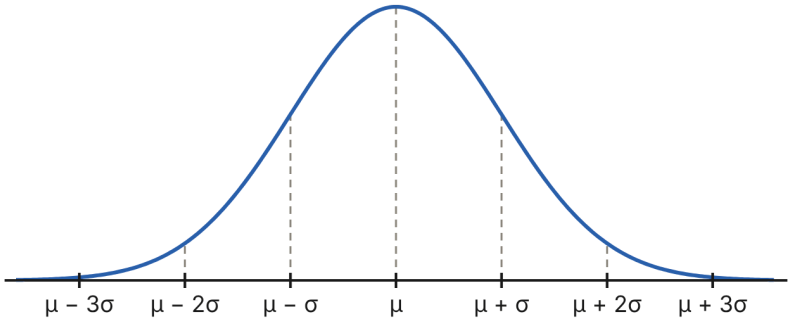
What percent of the area under the normal curve is shaded?



### Example 2

A normal distribution has mean μ and standard deviation σ. Find each probability.

a. $P(x \le \mu - \sigma)$ b. $P(x \ge \mu + 2\sigma)$



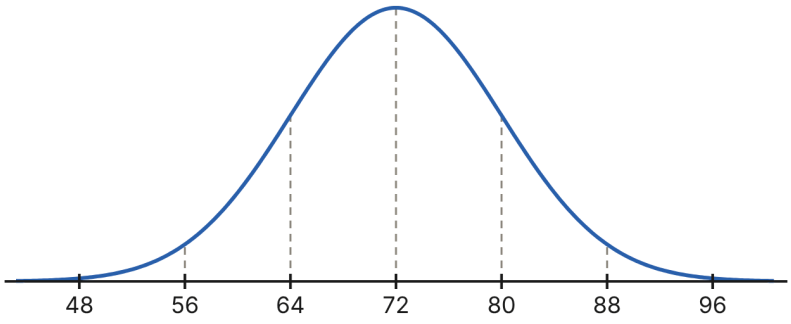

### Example 3

Test scores are normally distributed with a mean of 72 and a standard deviation of 8.

a. What percent of the scores are between 64 and 80?

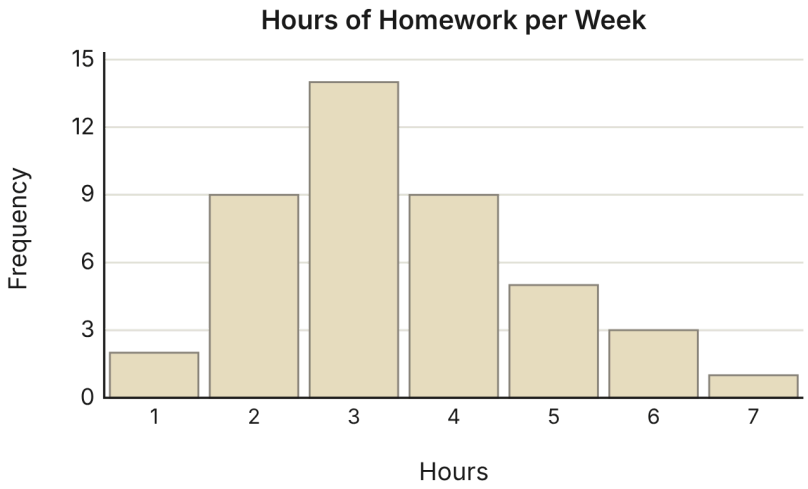
b. What percent are greater than 88?

c. In a group of 500 students, about how many score greater than 88?



**Example 4**

Does the histogram show data that are approximately normal? Is the mean greater or less than the median?




## Part 4 · Partner practice

**Partner 1**

A normal distribution has a mean of 150 and a standard deviation of 10. Find P(x < 130).

**Partner 2**

What percent of the area under a normal curve is between $\mu - 3\sigma$ and $\mu - \sigma$?

